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Studierichting: Informatica

Oefeningenreeks 2D nr. 11.4

$$\begin{aligned}\lim_{x \rightarrow 7} \left(\frac{3 - \sqrt{x+2}}{x-7} \right) &= \lim_{x \rightarrow 7} \left(\frac{(3 - \sqrt{x+2})(3 + \sqrt{x+2})}{(x-7)(3 + \sqrt{x+2})} \right) = \lim_{x \rightarrow 7} \left(\frac{9 - x - 2}{(x-7)(3 + \sqrt{x+2})} \right) \\ &= \lim_{x \rightarrow 7} \left(\frac{-(x-7)}{(x-7)(3 + \sqrt{x+2})} \right) = \lim_{x \rightarrow 7} \left(\frac{-1}{3 + \sqrt{x+2}} \right) = \frac{-1}{3 + \sqrt{9}} = -\frac{1}{6}\end{aligned}$$

References